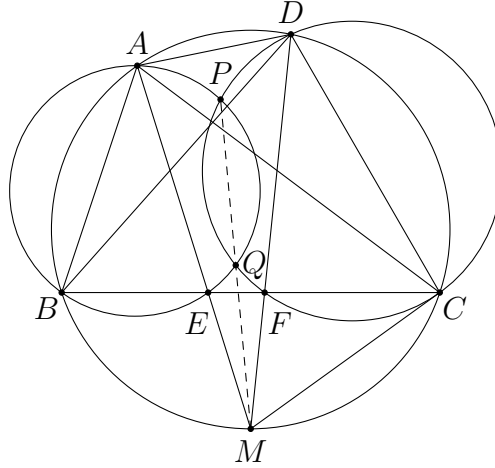


Radical Axes - Solutions

Problem 1. Let $ABCD$ be a cyclic quadrilateral. The internal angle bisector of $\angle BAC$ meets BC at E . The internal angle bisector of $\angle BDC$ meets BC at F . The circumcircles of $\triangle ABE$ and $\triangle CDF$ meet at two distinct points P and Q . Prove that AE , DF and PQ are concurrent.

Solution.

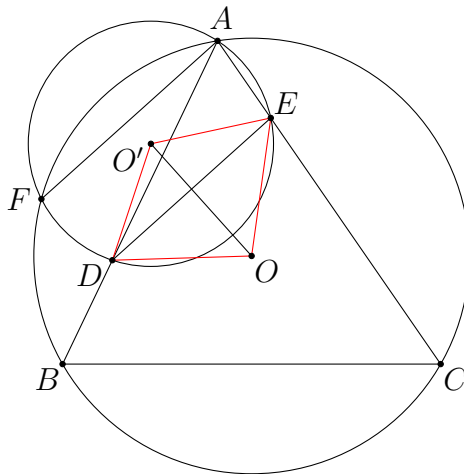


Note that AE and DF meet at the midpoint of the arc BC not containing A, D . From

$\angle MEF = \angle ACB + \angle MAC = \angle ACB + \angle BAM = \angle ACB + \angle BCM = \angle ACM = \angle ADM$,
points D, A, E, F are concyclic. So $MA \times ME = MD \times MF$. This shows M lies on the radical axis PQ of (ABE) and (CDF) as desired.

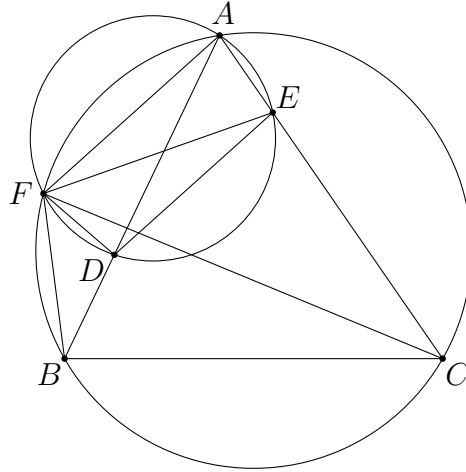
Problem 2. In $\triangle ABC$, the points D and E lie on the sides AB and AC respectively such that $AD \times DB = AE \times EC$. The circumcircles of $\triangle ABC$ and $\triangle ADE$ meet again at F . Prove that $AF \parallel DE$.

Solution 1.



Let O and O' be the circumcentres of $\triangle ABC$ and $\triangle ADE$ respectively. From the given condition $AD \times DB = AE \times EC$, the powers of D and E with respect to (ABC) are equal. So $OD = OE$. Together with $O'D = O'E$, we find that $DOEO'$ is a kite. So $DE \perp OO'$. As the radical axis AF of (ABC) and (ADE) is perpendicular to the line OO' , we have $AF \parallel DE$.

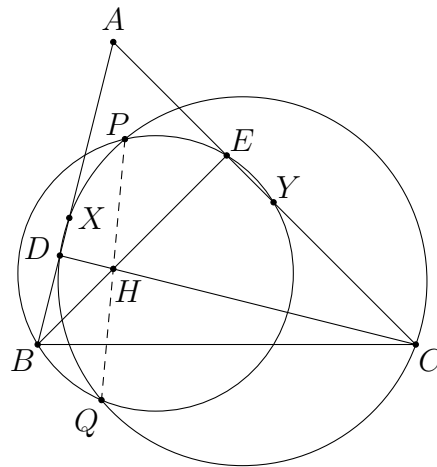
Solution 2.



By spiral similarity, we have $\triangle FBD \sim \triangle FCE$. This gives $\frac{FD}{FE} = \frac{BD}{CE} = \frac{AE}{AD}$. Together with $\angle DFE = \angle EAD$, we get $\triangle FDE \sim \triangle AED$. So $\angle FDE = \angle AED$, and hence $AF \parallel DE$ since $AFDE$ is cyclic.

Problem 3. Given $\triangle ABC$, two variable points X and Y lie on the sides AB and AC respectively. Suppose the circles with diameters BY and CX respectively intersect at two distinct points P and Q . Prove that the line PQ passes through a fixed point when X and Y vary.

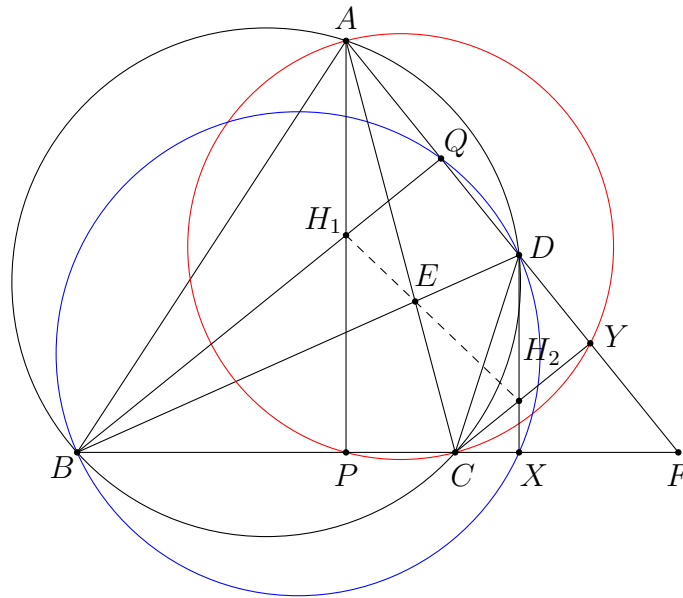
Solution.



We claim that the orthocentre H of $\triangle ABC$ is the fixed point. Let D and E be the feet of the altitudes from C and B respectively. Since $\angle CDX = \angle BEY = 90^\circ$, points D and E lie on the circles with diameters CX and BY respectively. As D, B, C, E are concyclic, we have $HB \times HE = HC \times HD$. So the powers of H with respect to (YEB) and (XDC) are equal. Hence, H lies on the radical axis PQ of the two circles as desired.

Problem 4. Let $ABCD$ be a cyclic quadrilateral. The diagonals AC and BD meet at E . The extension of the lines AD and BC meet at F . Let H_1 and H_2 be the orthocentres of $\triangle ABF$ and $\triangle CDF$ respectively. Prove that E, H_1 and H_2 are collinear.

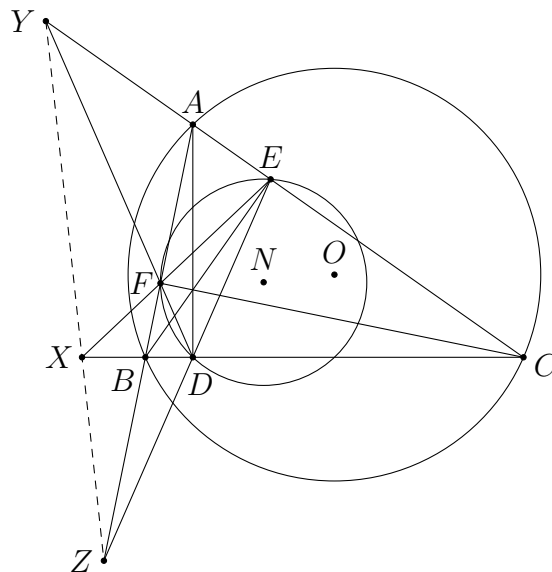
Solution.



Let the feet of the perpendiculars from A and D to BF be P and X respectively. Let the feet of the perpendiculars from B and C to AF be Q and Y respectively. As $\angle APC = \angle AYC = 90^\circ$, points A, P, C, Y are concyclic. Similarly, B, X, D, Q are concyclic. Also, A, B, P, Q are concyclic. This implies $H_1A \times H_1P = H_1B \times H_1Q$. Thus, H_1 lies on the radical axis of $(APCY)$ and $(BXDQ)$. Similarly, D, C, X, Y are concyclic, so that H_2 lies on the radical axis of $(APCY)$ and $(BXDQ)$. Lastly, from $EA \times EC = EB \times ED$, the point E also lies on this radical axis. So E, H_1, H_2 are collinear.

Problem 5. In $\triangle ABC$, let D, E, F be the feet of the altitudes from A, B, C respectively. The lines BC and EF meet at X . The lines CA and FD meet at Y . The lines AB and DE meet at Z . Prove that X, Y, Z lie on a line perpendicular to the Euler line of $\triangle ABC$.

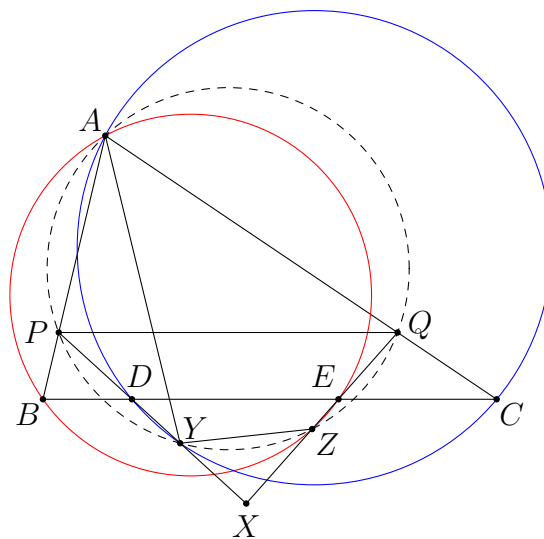
Solution.



Let O and N be the circumcentre and nine-point centre of $\triangle ABC$ respectively. As B, C, E, F are concyclic, we have $XB \times XC = XF \times XE$. This shows X lies on the radical axis of the circumcircle and nine-point circle of $\triangle ABC$. The same holds for Y and Z due to symmetry. So X, Y, Z are collinear, and they lie on a line perpendicular to the Euler line ON of $\triangle ABC$.

Problem 6. In $\triangle ABC$, let D and E be two points on the side BC . Let P and Q be points on the sides AB and AC respectively. The lines DP and EQ meet at X . Prove that X lies on the radical axis of the circumcircles of $\triangle ABE$ and $\triangle ACD$ if and only if $PQ \parallel BC$.

Solution.



Suppose the line DP meets (ACD) again at Y , and the line EQ meets (ABE) again at Z . If $PQ \parallel BC$, we have

$$\angle YPQ = \angle YDE = \angle YDC = \angle YAC = \angle YAQ.$$

This implies A, P, Y, Q are concyclic. Similarly, A, P, Z, Q are concyclic. Combining these, we find that A, P, Y, Z, Q are concyclic. Hence, we get $\angle XDE = \angle XPQ = \angle XZY$. It follows that D, Y, Z, E are concyclic, and hence $XD \times XY = XE \times XZ$. This implies X lies on the radical axis of $(ACYD)$ and $(ABZE)$.

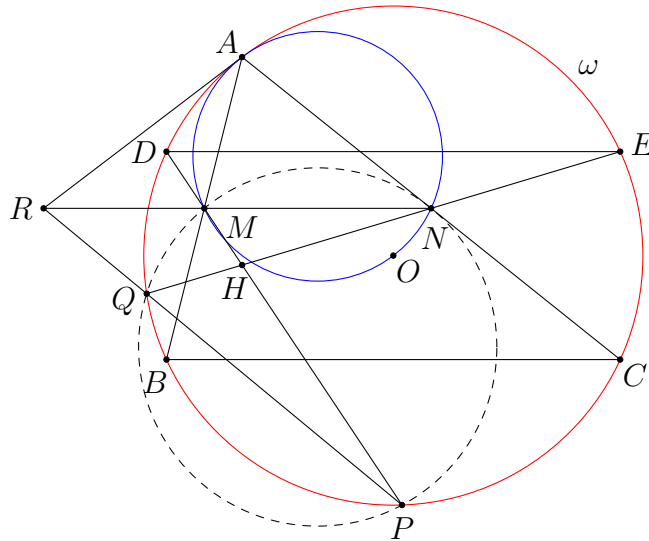
Conversely, if X lies on the radical axis of $(ACYD)$ and $(ABZE)$, then $XD \times XY = XE \times XZ$ and hence D, Y, Z, E are concyclic. Therefore, we have

$$\angle YAQ = \angle YAC = \angle YDC = \angle YDE = 180^\circ - \angle EZY = 180^\circ - \angle QZY.$$

This implies A, Y, Z, Q are concyclic. Similarly, A, P, Y, Z are concyclic. Combining these, we find that A, P, Y, Z, Q are concyclic. So $\angle YPQ = 180^\circ - \angle QZY = \angle YDE$. It follows that $PQ \parallel BC$.

Problem 7. (USA TSTST 2011 Problem 4) Acute triangle ABC is inscribed in circle ω . Let H and O denote its orthocentre and circumcentre, respectively. Let M and N be the midpoints of sides AB and AC , respectively. Rays MH and NH meet ω at P and Q , respectively. Lines MN and PQ meet at R . Prove that $OA \perp RA$.

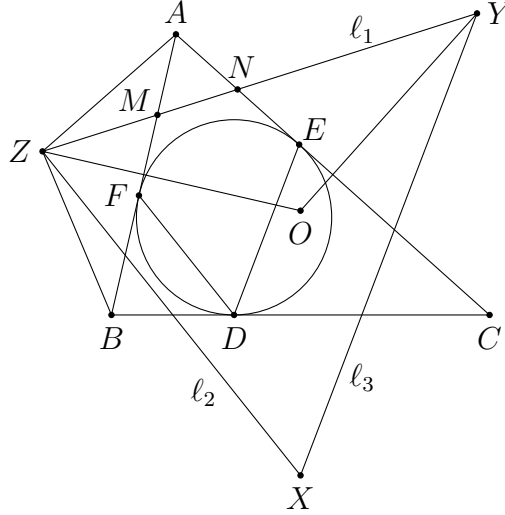
Solution.



Let MH meet ω again at D , and NH meet ω again at E . Recall that M and N are the midpoints of HD and HE respectively. So $MN \parallel DE$. It follows that $\angle PMN = \angle PDE = \angle PQE$. So N, M, Q, P are concyclic. Consider the radical centre of ω , $(NMQP)$ and (AMN) . Note that ω and (AMN) are tangent at A as $\triangle AMN$ and $\triangle ABC$ are homothetic. So the radical centre is the concurrent point of the lines PQ , MN and the tangent to ω at A . This means RA is tangent to ω , so that $OA \perp RA$.

Problem 8. The incircle of $\triangle ABC$ touches the sides BC, CA, AB at points D, E, F respectively. Let ℓ_1 be the line equidistant to point A and the line EF , ℓ_2 the line equidistant to point B and the line FD , and ℓ_3 the line equidistant to point C and the line DE . Prove that the circumcentre of the triangle bounded by ℓ_1, ℓ_2 and ℓ_3 coincides with the circumcentre of $\triangle ABC$.

Solution.



Let ℓ_1 meet ℓ_2 at Z , ℓ_2 meet ℓ_3 at X , ℓ_3 meet ℓ_1 at Y . Let M and N be the midpoints of AF and AE respectively. Since $MA^2 = MF^2$, M lies on the radical axis of the degenerated circle A and the incircle. The same holds for N by symmetry. So ℓ_1 is the radical axis. Thus, Z lies on this radical axis. Similarly, Z lies on the radical axis of B and the incircle. It follows that Z is the radical centre of A, B and the incircle. This implies $ZA = ZB$.

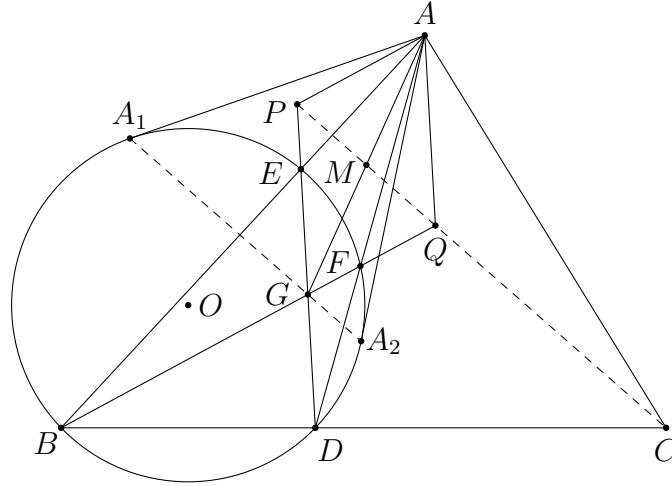
Let O be the circumcentre of $\triangle ABC$. Then $OA = OB$. Thus, $AZBO$ is a kite, and we have $AB \perp OZ$. Similarly, $AC \perp OY$. Therefore, we find that $\angle YOZ = 180^\circ - A$. Also, as $XY \parallel DE$ and $XZ \parallel DF$, we have

$$\angle YXZ = \angle EDF = 180^\circ - \angle CDE - \angle FDB = 180^\circ - \left(90^\circ - \frac{C}{2}\right) - \left(90^\circ - \frac{B}{2}\right) = \frac{B+C}{2}.$$

This implies $2\angle YXZ = \angle YOZ$. Similarly, we have $2\angle ZYX = \angle ZOY$. These show O is the circumcentre of $\triangle XYZ$.

Problem 9. (China TST 2007 Problem 1) Let D be a point on the side BC of $\triangle ABC$ such that $\angle CAD = \angle CBA$. Circle O passing through the points B, D intersects the segments AB, AD at E, F respectively. BF and DE intersect at G . M is the midpoint of AG . Prove that $CM \perp AO$.

Solution. Firstly, from $\angle DAC = \angle CBA$, we know that CA is a tangent to (ABD) . Therefore, we have $CA^2 = CB \times CD$. This shows C lies on the radical axis of the degenerated circle A and circle O . If we can show that M lies on this radical axis, then $CM \perp AO$.



One way is to let A_1 and A_2 be the contact points of the tangents from A to circle O . It is well-known that A_1, G, A_2 lie on the polar of A with respect to circle O , and hence they are collinear. The radical axis of A and circle O is the line equidistant to A and the line A_1A_2 . So M lies on this radical axis as desired.

Alternatively, Suppose the radical axis of A and circle O meets the lines DE and BF at P and Q respectively. Then $PA^2 = PE \times PD$, yielding $\angle PAE = \angle ADE = \angle FBE$. This implies $PA \parallel BF$. Similarly, we have $QA \parallel DE$. So $APGQ$ is a parallelogram. It follows that the midpoint M of AG lies on PQ , which is the radical axis of A and circle O as desired.

Problem 10. In $\triangle ABC$, the internal angle bisector of $\angle BAC$ meets the side BC at D . The common external tangent of the circumcircles of $\triangle ABD$ and $\triangle ACD$ which is closer to D touches the two circles at E and F respectively. Let P be the point on the ray AD such that $\triangle ABP \sim \triangle APC$. Prove that $DEPF$ is a parallelogram.

Solution. Let O_1 and O_2 be the circumcentres of (ABD) and (ACD) respectively. Let X be the external homothetic centre of the two circles, i.e. X is the intersection of EF and O_1O_2 . As

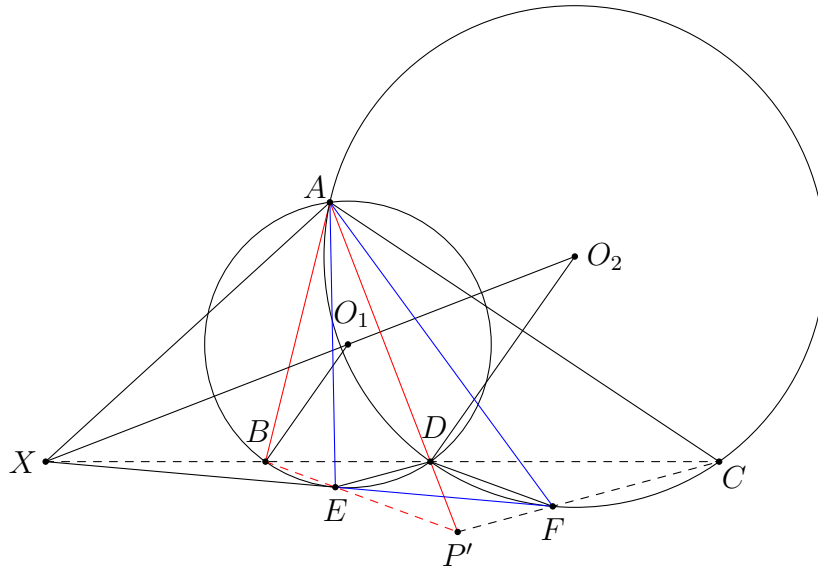
$$\angle CDO_2 = 90^\circ - \angle DAC = 90^\circ - \angle BAD = \angle DBO_1,$$

we have $DO_2 \parallel BO_1$. This implies X, B, D are collinear, i.e. X lies on BC .

The homothety mapping (ABD) to (ACD) maps the points B, D, E to the points D, C, F respectively. So we have $BE \parallel DF$. This yields

$$\angle FEB = 180^\circ - \angle DFE = 180^\circ - \angle DCF.$$

Thus, C, B, E, F are concyclic. By considering the radical centre of (ABD) , (ACD) and $(CBEF)$, the lines BE, CF, AD are concurrent at a point P' . We claim that $P' = P$.



As EF is tangent to (ABE) , we have $\angle FEA = \angle P'BA$. Also, we find that

$$\angle DAF = \angle DFE = \angle BEX = \angle BAE.$$

So $\angle EAF = \angle EAD + \angle DAF = \angle EAD + \angle BAE = \angle BAP'$. These imply $\triangle AEF \sim \triangle ABP'$. By symmetry, we also have $\triangle AEF \sim \triangle AP'C$. So P' is the unique point on AD such that $\triangle ABP' \sim \triangle AP'C$ (the uniqueness comes from the fact that this is equivalent to $AP'^2 = AB \times AC$). Thus, $P' = P$. Then we have

$$\angle FPD = \angle CPA = \angle PBA = \angle EBA = \angle EDP.$$

This shows $PF \parallel ED$. Similarly, $PE \parallel FD$. As a result, $DEPF$ is a parallelogram.